

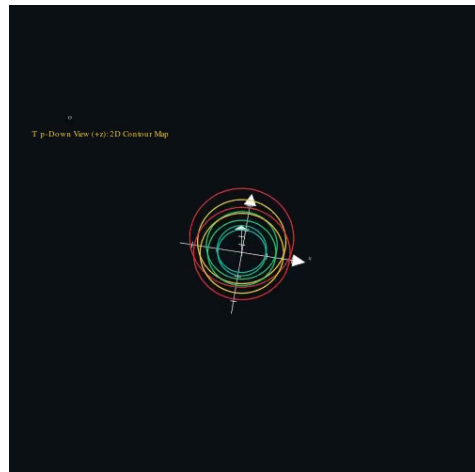
Level Curves: 3D Surface to 2D Contour Map

From 3D Slices to 2D Projections

- **In 3D Space:** Slicing the surface $z = f(x, y)$ by horizontal planes $z = k_1, k_2, \dots$ produces 3D space curves.
- **Vertical Projection:** Projecting each 3D curve vertically downward onto the xy -plane ($z = 0$) gives the **Level Curves**:

$$L_k = \{(x, y) \in \mathbb{R}^2 \mid f(x, y) = k\}.$$

- **Top-Down View (+z):** Looking directly down from the z -axis reveals the complete **2D Contour Map**.



Smooth transition: 3D Slicing $\longrightarrow$ Top-Down (+z) Contour Map